



Ryan Zernach Full-Stack AI Systems Engineer • Agentic Infrastructure & Product Engineering

Portfolio: [Ryan.Zernach.com](#) • Email: Ryan@Zernach.com • LinkedIn • GitHub • Remote • Healthcare • FinTech

HIGHLIGHTS

- Distributed systems engineer with 10+ years of engineering Back-End Infra, Agents, DevOps, Mobile Apps, Websites, & MLOps
- Scaled CellarChat™ tool-calling agent to 1M+ active members, complete with RLHF & CI/CD LLM evaluation scoring pipelines
- Engineered Apple Pay subscriptions to lift recurring revenue by 25%+ which lifted mobile app ratings to 4.9★ (10k+ reviews)
- Completed Gauntlet AI engineering fellowship with 2% acceptance rate and mastered the art of agentic AI orchestration
- Designed observable backend systems that support autonomous AI-powered VoIP eCommerce payment automations
- Awarded Top 10 of 10,000 international projects in 2024 NASA Space Apps hackathon with AI Voice Assistant

TECH STACK

Programming Languages
TypeScript, Python, Rust, Golang, Java, C#, Ruby, Swift, Kotlin, Bash, JavaScript, PHP C++, Objective-C, R, Solidity, Dart, Flutter, XML

AI Systems & Machine Learning
LLaMA, RAG, Agents, MCP Evaluations, Semantic Kernel, LangChain, Langfuse, ML Pipelines, Computer Vision, LlamaLM, Vector Embeddings, PyTorch, Knowledge Graphs, TensorFlow, Scikit-Learn, Bedrock, Foundry

Platform & Backend Engineering
DevEx, Node.js, FastAPI, Django, Next.js, Express, .NET Core, OpenAPI, Microservices, REST, GraphQL, WebSockets, Webhooks, gRPC, Kafka

Data & Persistence
PostgreSQL, SQL, NoSQL, Redis, Vector Databases, ETL, Snowflake, Graph Databases, Entity Relationships, Caching, BigQuery, PySpark

Cloud Infrastructure & DevOps
AWS, Azure, GCP, Cloudflare, Docker, GitHub Actions, Terraform, Kubernetes, CI/CD, CloudOps, Blue/Green, DNS, Networking

Front-End & Product Engineering
React, React Native, iOS, Android, Web, Swift, Kotlin, Expo Router, Next.js, Redux, Streamlit, Animations, Design, User Feedback, Playwright

Distributed Systems & Operations
Queues, Redis, RabbitMQ, Analytics, Playbooks, Linux, Airflow, Event-Driven Architecture, VoIP, Observability, Admin Systems, WebRTC, Rate Limiting, Horizontal Scaling, Load Balancing, Cron

Security, Identity & Compliance
AuthN, AuthZ, OAuth, SOC 2, Identity, Role-Based Access Control, JWT, Secure Systems Design, SSO, API Security, Secrets Management

Payments, Communications & Integrations
Stripe, Apple Pay, iAP Billing, Subscriptions, Webhooks, Twilio, SMS, Push Notifications, HTTP APIs, Marketing Tech, Email SMTP/IMAP

Blockchain & Web3
Ethereum, Solana, dApps, WalletConnect, MetaMask, Polygon, Solidity, Smart Contracts, Ethers.js, Tokens, ERC-20, ERC-721

EXPERIENCE

Independent Full-Stack AI Systems Engineer • USA • Remote

July 2025 - Present

- Building AI-enabled products and back-end systems across fintech, semantic intelligence, and now focused on healthcare
- Productionized eCommerce platform powered by Twilio VoIP + real-time voice models to automate outbound phone calls
- Published TranslateWallpaper, a cross-platform mobile app that creates LLM-powered language-learning wallpapers
- Dockerized agentic operating-system around local hardware inferences, MCP, OSS, Plaid & federal tax preparations

Senior Full-Stack AI Systems Engineer • CellarTracker • Seattle • Remote

Aug 2023 - Jul 2025

- Scaled agentic CellaChat™ to 1M+ international members & released world's highest-rated #1 wine app
- Architected OpenAPI pipeline, embeddings, fine-tuning, LLM evaluations, RLHF, and admin observability
- Integrated intelligent ML-powered recommendations for world's largest online database of wine reviews
- Maintained Python, TypeScript, React Native, Redis, C#/NET, Microsoft Azure AI Foundry, & DevOps CI/CD
- Helped to boost app store ratings to 4.9★ average with 10,000+ verified 5-star ratings on iOS and Android
- Embedded Apple Pay subscriptions in the React Native app to generate additional 25%+ recurring revenue
- Developed backend Python systems supporting agentic workflows, support tickets, & account management
- Improved user's experience of cellar management CRUD workflows to improve consistency and reduce taps
- Implemented animations, barcode scanning, activity feed posts, support ticket refactors, & OCR image search
- Streamlined the Redux store for fast, efficient, and persistent global state management between app sessions
- Migrated legacy support ticket system from Active Server Pages (ASP) to safer, type-friendly Express framework
- Rebuilt the React Native app's analytics implementations for tracking critical user interactions and in-app behavior
- Arranged A/B tests during payroll & onboarding experiences using phased rollout schedules for marketing analysis

Full-Stack Mobile Engineer • Archlife Industries Software • Miami • Remote

Sept 2021 - Aug 2023

- Delivered eTrading exchange for stocks/crypto, including public leaderboards and copy friend's portfolio distribution
- Used AWS Terraform to reliably deploy Java with Springboot, docker images, and admin observability dashboard
- Constructed core mobile app functionality, including: Plaid connections, trending investors, & suggested stocks
- Built social network features, such as direct chat, user profiles, posts, likes, emojis, & threaded comments
- Wrote the data-visualization logic for viewing and comparing the performance of different stocks/crypto
- Transitioned eSports React Native codebase from JavaScript to TypeScript for long-term maintainability
- Improved Ruby on Rails for push notifications, chat, & native calendar sync to improve user satisfaction
- Remedied bluetooth connection & map performance for Swift iOS/iPad telecommunications app

Data Science & Machine Learning Tutor • edX • Vanderbilt, UC Berkeley, +24 others • Remote

July 2020 - Sept 2021

- During pandemic, taught 900+ online college students Python, back-end, machine learning, TypeScript, websites, & databases
- Managed busy schedule of remotely teaching 30+ university students per week in 1:1 setting (one hour per person)
- Dozens of universities were subscribed to my employer's curriculum and online tutoring resources for students

React Web Developer (Full-Stack) • Georgia Hybrid Landscaping • Alpharetta, Georgia

Oct 2015 - Jan 2019

- Engineered uploading videos/images, survey questions, pricing proposal, & processing payments via Java Spring Boot
- Migrated websites from PHP to TypeScript React, which successfully processed \$500k+ gross sales in three years
- Managed online eCommerce payments, visitor analytics, Google AdSense, and email marketing campaigns
- Configured outbound marketing email campaign schedulers & listeners using IMAP/POP
- Setup observability dashboards for payments, marketing, analytics, API logs, & admin

Payments Systems Intern (Card Processing) • Cost Plus Processing • Alpharetta, Georgia

Sept 2012 - Jan 2013

- Worked with merchant processing systems & payment gateways to understand critical financial flows
- Analyzed end-to-end card transaction lifecycles (interchange rates, risk) across payment networks
- Built foundational knowledge of payment infrastructure, interchange fees, & transaction routing

EDUCATION & TRAINING

AI Engineering Fellowship (G3) • Gauntlet AI • Remote • Austin, TX

2025

- Mastered how to command agents, automate workflows, and 100x enterprise-grade software engineering
- Harnessed the power of RAG, multi-agent systems, OpenAPI, and MCP to build agentic automations

Data Science & Machine Learning Bootcamp • Lambda School • Y Combinator • Remote

2020

- Recommendation Engines, Computer Vision (CV) Models, Time-Series Forecasting, Natural-Language Processing
- Python, Scikit-Learn, PySpark, Django, Flask, FastAPI, Data Engineering, PostgreSQL Databases, Back-End APIs

Turfgrass & Landscape Management • University of Georgia (NALP) • Athens, Georgia

2015

- Design Principles, Science of Grass, Outdoor Business Management, Landscape Architecture

Computer Science & Mathematics Coursework • Georgia Southern University • Statesboro, Georgia

2012

- Visual Basic, Calculus, C, C++, R, Statistics, Data Structures, Programming Principles, Linear Regression

CERTIFICATIONS

2025 • Gauntlet AI Engineering Fellowship (G3)

2024 • NASA Space Apps Hackathon Finalist

2022 • Blockchain Hackathon Participant

2020 • TensorFlow Computer Vision

2020 • AutoML Pipelines (DataRobot Academy)

2019 • Google Cloud Platform GCP Developer

2019 • AWS Microservices Cloud Practitioner

2019 • Y Combinator Startup School

RYAN ZERNACH
Full-Stack AI Systems Engineer



BUILDING RELIABLE
HEALTHCARE AI

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Healthcare Technology is My #1 Priority

I have built production AI systems used by 1M+ international members. Now I want to build where the work has real human stakes: healthcare. I am drawn to systems that help clinicians move faster, think clearer, and trust the tools in front of them when pressure is high. If your company is building serious AI for real healthcare workflows, I would be honored to help.

RELATED LINKS

[Production LLM Systems](#) [AI Automations](#) [OSS Contributions](#)



HIPAA-Compliance is an Engineering Discipline

In healthcare AI, I treat HIPAA as a system design requirement, not a compliance checkbox. My default is minimum-necessary data flow, strict access boundaries, and auditability from day one so ePHI stays protected without slowing clinicians down. I build for the operational realities that matter in production: least-privilege access, controlled change management, incident-ready processes, and evidence-backed controls that hold up in review. My fintech work at Finequities sharpened those habits, and I apply them directly to HIPAA-heavy environments where privacy intent has to show up in everyday engineering execution.

- Data handling boundaries: explicit separation of ePHI-bearing systems, scoped interfaces, and minimized replication paths.
- Identity and access controls: role-based access, least privilege, and strict production permission lifecycle management.
- Encryption and transport safety: protected data in transit and at rest, with key-management decisions tied to risk level.
- Audit trail integrity: immutable access and change logs that support internal reviews, incident analysis, and compliance evidence.
- Vendor and integration guardrails: third-party review standards, BAA-aware decisions, and no silent data leakage through tooling.
- Incident preparedness: playbooks, escalation paths, and post-incident correction loops with real ownership.
- Release discipline: compliance-sensitive changes gated by clear review criteria before production deployment.



RELATED LINKS

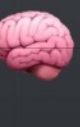
[Finequities SOC 2 Compliance](#)



Why Healthcare?

Healthcare is where great engineering meets human consequence, and it is the one industry every human on earth depends on many times in life. I have spent years building scalable AI in high-pressure production environments, and I want to channel that same operating model into healthcare with humility and care. In most industries, better software improves convenience. In healthcare, better systems can return time to clinicians, reduce preventable friction, and support clearer decisions when stakes are high. That is the challenge I want most: complex work, high standards, and real accountability to outcomes. I am drawn to environments where reliability is non-negotiable, trust must be earned every day, and technical quality is measured by how well the system serves people under pressure.

This path is also personal for me. My sister works in nursing, and healthcare has been a serious part of my family story. I have seen how demanding this work is and how much impact better tools can have for people on the front lines. Healthcare runs in the family, and I want to contribute from my side of the table by building AI systems that are dependable, intelligent, and truly worthy of the people who rely on them. My passion is to build technology that gives clinicians more support and gives patients better chances at healthier, happier, longer lives.



Mindset Matters Most

My engineering style is ambitious, disciplined, and production-minded, but always in service of the people who depend on the system. I want AI to stand out for the right reasons: it works, it is understandable, it is dependable, and it earns trust when the pressure is real.

- Start from the real workflow, the real operators, and the real failure modes, then build around safety, observability, and crisp ownership boundaries.
- Treat quality gates as core infrastructure, not cleanup work somebody promises to do later.
- Ship meaningful progress quickly, then harden reliability, controls, and clarity every cycle.
- Build systems that teams can trust, understand, and improve long after the first release.

RELATED LINKS

[Hardware is Horsepower](#) [HIPAA Architecture](#) [AI Engineering Methodologies](#)



Healthcare Career Goals

- Full-Stack AI Systems Engineer roles building healthcare technology that visibly improves patient outcomes, clinician effectiveness, access to care, and the dignity of the care experience.
- Teams that treat AI evaluation, monitoring, privacy, safety, and deployment discipline as foundational engineering responsibilities, especially when the product touches high-stakes clinical or patient workflows.
- Opportunities to apply my experience in LLM infrastructure, RAG, agents, evaluations, distributed systems, mobile apps, and cloud operations to healthcare systems where reliability and human impact both matter.
- Products that help clinicians spend less time fighting fragmented software and more time exercising judgment, empathy, and care.
- High-standard environments where rigor beats hype, evidence beats demos, and the mission is worthy of our effort.



Closing Note

I have built scalable AI systems for large, real-world audiences, and I want my next chapter to be healthcare with the same seriousness and deeper purpose. I care deeply about helping people live healthier, happier, longer lives, and I want that mission to define what I build next. If your team is building healthcare AI with real standards, real ambition, and a real commitment to quality, I would value the chance to talk. I am ready to help build systems people remember quietly, for the best reason: they worked when they were needed most.

SCHEDULE A MEETING

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